

BINCOM ACADEMY

Cybersecurity

Network Attack Detection and Reporting

Assignment 1: Cowrie Honeypot Security Monitoring Lab

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Institution: BINCOM Academy

Course: Cybersecurity

Assignment: Assignment 1

Date: 7 October 2026

Tools: Kali Linux, Ubuntu, Cowrie, Nmap, Wireshark, OpenSSH

INCIDENT REPORT

Incident ID	IR-2026-1007-01	Date reported	7 October 2026
Reported by	Akinrelere Oluwatobi (BINCOM Academy)	Status	Closed (controlled lab exercise)
Category	Network reconnaissance and simulated unauthorised SSH access	Severity	Informational (lab, no production asset)
Detection sources	Cowrie honeypot logs, Wireshark capture, Nmap output	Affected host	Cowrie honeypot, Ubuntu 10.0.2.15, TCP 2222 (SSH)

1. Summary

A Cowrie SSH honeypot (Ubuntu, 10.0.2.15) detected activity from a second lab host, Kali Linux (10.0.2.11): a service-discovery scan followed by a direct SSH connection to the honeypot on TCP 2222. An earlier local test session on the honeypot also logged a root login and three shell commands. Wireshark independently captured the TCP handshake and SSH key exchange, and the source IP and source port (40294) match the Cowrie session record. All activity was generated by the author in an isolated VirtualBox network. No real system was compromised.

2. Timeline (Cowrie log timestamps)

Time	Event
20:24:36	Cowrie started; SSH listener active on TCP 2222.
20:34:52 to 20:40:07	Local test session 10d0ac801daa (127.0.0.1): root login accepted by the honeypot, commands whoami , pwd , ls run in the emulated shell; TTY recording saved; session closed after 271.7 s.
20:58:52	Cowrie stopped and restarted (configuration re-read from etc/cowrie.cfg).
20:59:30	Nmap -sV probe from 10.0.2.11:38578 (session e4b9a9b04144); connection closed after 63 ms. Nmap reported 2222/tcp open ssh .
21:06:04	SSH connection from 10.0.2.11:40294 (session 63f7a45ac1bf). Client banner and HASSH fingerprint logged. Same port seen in Wireshark packets 9 to 11 (SYN, SYN/ACK, ACK).
21:08:07	Honeypot timeout reached; session closed after 123.34 s. No successful login logged for this session.

3. Indicators observed

Indicator	Value
Source IP / MAC	10.0.2.11 / 08:00:27:8a:35:d2 (Kali Linux VM)
Source ports	38578 (Nmap probe), 40294 (SSH session)
Destination	10.0.2.15, TCP 2222 (Cowrie SSH)
SSH client banner	SSH-2.0-OpenSSH_10.4p1 Debian-4
HASSH fingerprint	eeeca2460550b9ded084ecf2f70a75356
Session IDs	63f7a45ac1bf (SSH), e4b9a9b04144 (Nmap), 10d0ac801daa (local test)

4. Analysis and impact

The pattern (port scan, then SSH connection, then discovery commands such as whoami and ls) mirrors early-stage attacker behaviour: MITRE ATT&CK T1046 Network Service Discovery, T1021.004 Remote Services: SSH, T1033 System Owner/User Discovery and T1083 File and Directory Discovery. The honeypot presents a fake Debian 12 / OpenSSH 9.2p1 identity, so the scan result reflects the emulation, not the real Ubuntu host. Impact: none. The shell is emulated, no files were transferred and no production system was reachable.

5. Response and recommendations

(1) Forward honeypot and network logs to a SIEM such as Wazuh or Splunk. (2) Alert on any connection to the honeypot, since no legitimate user should touch it. (3) Block or rate-limit repeat sources and track HASSH values to recognise recurring clients. (4) Keep the honeypot isolated from production networks. (5) Preserve the packet capture and Cowrie logs as evidence. Supporting screenshots are in Figures 1 to 11 of this report.

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Page 2 of this document is the stand-alone one-page incident report required by the assignment.

1. Executive Summary

This report documents a controlled network attack detection exercise carried out for Assignment 1. Two virtual machines were built in VirtualBox on the same NAT network: a Kali Linux machine (10.0.2.11) used as the scanning and testing system, and an Ubuntu machine (10.0.2.15) running the Cowrie SSH honeypot on TCP port 2222.

Nmap service detection from Kali identified the honeypot's SSH service. A direct SSH connection from Kali was then made to the honeypot and captured in Wireshark, showing the TCP three-way handshake, the SSH banner exchange and the key-exchange negotiation. Cowrie independently logged the same connection, including the source IP, source port, SSH client banner, HASSH fingerprint, session ID and duration. A separate local test login on the honeypot also produced logged shell commands and a TTY recording.

Matching the Wireshark packets to the Cowrie record by IP address and source port (40294) shows how network captures and honeypot telemetry reinforce each other during an investigation.

2. Objectives

- Deploy Kali Linux and an Ubuntu host running the Cowrie honeypot on separate virtual machines.
- Use Nmap to discover the exposed service on the honeypot host.
- Generate controlled SSH activity against the honeypot, including a simulated unauthorised login and basic discovery commands.
- Capture the resulting traffic with Wireshark and analyse it.
- Examine Cowrie logs and session artefacts and correlate them with the packet capture.
- Produce a professional report and a one-page incident report.

3. Lab Environment

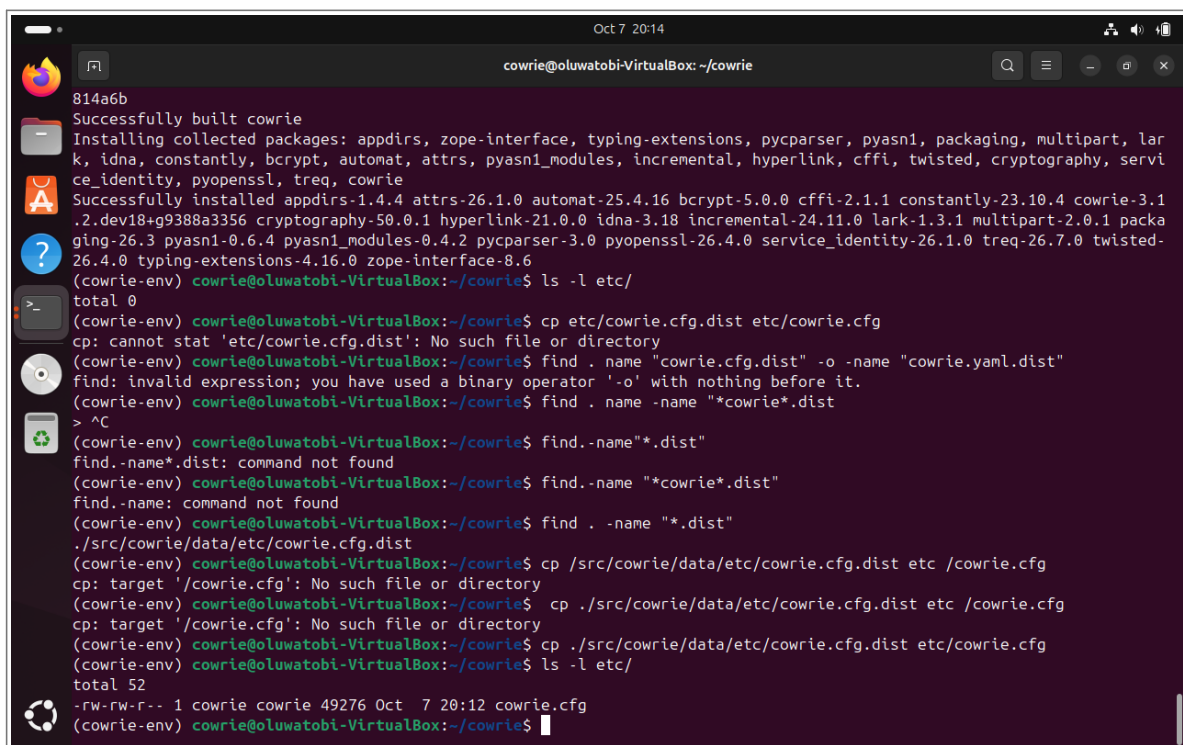
System	Role	Address / Port	Notes
Kali Linux VM	Scanning and testing system	10.0.2.11	Nmap 7.99, Wireshark, OpenSSH client
Ubuntu VM	Cowrie honeypot host	10.0.2.15	Ubuntu 24.04 (noble); Cowrie 3.1.2.dev18 in a Python 3.12 virtual environment under user <i>cowrie</i>
Cowrie SSH service	Honeypot service	TCP 2222	JSON log output enabled. A Telnet endpoint (TCP 2223) is defined in the configuration but only the SSH service was started

Both machines were attached to the same VirtualBox NAT network. Cowrie runs under a dedicated, password-disabled *cowrie* user rather than as root, which limits the damage if the honeypot itself were ever exploited.

4. Methodology and Evidence

4.1 Honeypot deployment

A dedicated *cowrie* user was created, the Cowrie repository was cloned from GitHub into its home directory, and a Python virtual environment was created and activated. Cowrie was installed into the environment with pip, and the default configuration file was copied from the package data directory to *etc/cowrie.cfg*. The listen endpoints were checked in the configuration. SSH listens on TCP 2222; a Telnet endpoint on TCP 2223 is also defined but was not used in this lab. Screenshots of the user creation, repository clone and virtual environment are in Appendix A.



```
Oct 7 20:14
cowrie@oluwatobi-VirtualBox: ~/cowrie
814a6b
Successfully built cowrie
Installing collected packages: appdirs, zope-interface, typing-extensions, pycparser, pyasn1, packaging, multipart, lar
k, idna, constantly, bcrypt, automat, attrs, pyasn1_modules, incremental, hyperlink, cffi, twisted, cryptography, servi
ce_identity, pyopenssl, treq, cowrie
Successfully installed appdirs-1.4.4 attrs-26.1.0 automat-25.4.16 bcrypt-5.0.0 cffi-2.1.1 constantly-23.10.4 cowrie-3.1
.2.dev18+g9388a3356 cryptography-50.0.1 hyperlink-21.0.0 idna-3.18 incremental-24.11.0 lark-1.3.1 multipart-2.0.1 packa
ging-26.3 pyasn1-0.6.4 pyasn1_modules-0.4.2 pycparser-3.0 pyopenssl-26.4.0 service_identity-26.1.0 treq-26.7.0 twisted-
26.4.0 typing-extensions-4.16.0 zope-interface-8.6
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ ls -l etc/
total 0
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cp etc/cowrie.cfg.dist etc/cowrie.cfg
cp: cannot stat 'etc/cowrie.cfg.dist': No such file or directory
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ find . name "cowrie.cfg.dist" -o -name "cowrie.yaml.dist"
find: invalid expression; you have used a binary operator '-o' with nothing before it.
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ find . name -name "*cowrie*.dist"
> ^C
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ find.-name "*.dist"
find.-name*.dist: command not found
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ find.-name "*cowrie*.dist"
find.-name: command not found
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ find . -name "*.dist"
./src/cowrie/data/etc/cowrie.cfg.dist
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cp ./src/cowrie/data/etc/cowrie.cfg.dist etc /cowrie.cfg
cp: target '/cowrie.cfg': No such file or directory
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cp ./src/cowrie/data/etc/cowrie.cfg.dist etc /cowrie.cfg
cp: target '/cowrie.cfg': No such file or directory
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cp ./src/cowrie/data/etc/cowrie.cfg.dist etc/cowrie.cfg
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ ls -l etc/
total 52
-rw-rw-r-- 1 cowrie cowrie 49276 Oct 7 20:12 cowrie.cfg
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$
```

Figure 1: Cowrie installed successfully (cowrie-3.1.2.dev18) and the configuration file *etc/cowrie.cfg* created.

Cowrie was then started and its status checked. The service reported that it was running.

```

Oct 7 20:26
cowrie@oluwatobi-VirtualBox: ~/cowrie

(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ ls -l etc/
total 52
-rw-rw-r-- 1 cowrie cowrie 49276 Oct  7 20:12 cowrie.cfg
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ nano ./etc/cowrie.cfg
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ grep "listen_endpoints" ./etc/cowrie.cfg
listen_endpoints = tcp:6415:interface=127.0.0.1
# (default: listen_endpoints = tcp:2222:interface=0.0.0.0)
# listen_endpoints = systemd:domain=INET:index=0
# For both IPv4 and IPv6: listen_endpoints = tcp6:2222:interface=\:
# e.g listen_endpoints = "tcp:2222:interface=0.0.0.0 tcp:1022:interface=0.0.0.0" will result listening both on ports 22
# the proxy's, e.g. listen_endpoints = haproxy:tcp:2222:interface=0.0.0.0
listen_endpoints = tcp:2222:interface=0.0.0.0
# (default: listen_endpoints = tcp:2223:interface=0.0.0.0)
# listen_endpoints = systemd:domain=INET:index=0
# For IPv4 and IPv6: listen_endpoints = tcp6:2223:interface=\: tcp:2223:interface=0.0.0.0
# e.g "listen_endpoints = tcp:2223:interface=0.0.0.0 tcp:2323:interface=0.0.0.0" will result listening both on ports 22
# listen_endpoints = haproxy:tcp:2223:interface=0.0.0.0
listen_endpoints = tcp:2223:interface=0.0.0.0
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ bin/cowrie start
-bash: bin/cowrie: No such file or directory
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ bin/cowrie start
-bash: bin/cowrie: No such file or directory
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cowrie start

Join the Cowrie community at: https://www.cowrie.org/slack/

Starting cowrie: [twistd --umask=0022 --pidfile var/run/cowrie.pid --logger cowrie.python.logfile.logger cowrie]...
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cowrie status
cowrie is running (PID: 9150).
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$

```

Figure 2: Honeypot running: cowrie start followed by cowrie status reporting "cowrie is running (PID: 9150)".

```

Oct 7 20:28
cowrie@oluwatobi-VirtualBox: ~/cowrie

# listen_endpoints = systemd:domain=INET:index=0
# For IPv4 and IPv6: listen_endpoints = tcp6:2223:interface=\: tcp:2223:interface=0.0.0.0
# e.g "listen_endpoints = tcp:2223:interface=0.0.0.0 tcp:2323:interface=0.0.0.0" will result listening both on ports 22
# listen_endpoints = haproxy:tcp:2223:interface=0.0.0.0
listen_endpoints = tcp:2223:interface=0.0.0.0
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ bin/cowrie start
-bash: bin/cowrie: No such file or directory
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ bin/cowrie start
-bash: bin/cowrie: No such file or directory
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cowrie start

Join the Cowrie community at: https://www.cowrie.org/slack/

Starting cowrie: [twistd --umask=0022 --pidfile var/run/cowrie.pid --logger cowrie.python.logfile.logger cowrie]...
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ cowrie status
cowrie is running (PID: 9150).
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ tail -f var/log/cowrie/cowrie.log
2026-10-07T20:24:36.341192Z [cowrie.plugin#info] Python Version 3.12.3 (main, Aug 31 2026, 10:18:26) [GCC 13.3.0]
2026-10-07T20:24:36.341221Z [cowrie.plugin#info] Twisted Version 26.4.0
2026-10-07T20:24:36.341231Z [cowrie.plugin#info] Cowrie Version 3.1.2.dev18+g9388a3356
2026-10-07T20:24:36.341238Z [cowrie.plugin#info] Sensor UUID: bb878cb5-c284-11f1-af75-0800278dc026
2026-10-07T20:24:36.343313Z [cowrie.core.output#info] Loaded output engine: jsonlog
2026-10-07T20:24:36.344928Z [twisted.scripts._twistd_unix.UnixAppLogger#info] twistd 26.4.0 (/home/cowrie/cowrie/cowrie
-env/bin/python 3.12.3) starting up.
2026-10-07T20:24:36.344999Z [twisted.scripts._twistd_unix.UnixAppLogger#info] reactor class: twisted.internet.epollreac
tor.EPollReactor.
2026-10-07T20:24:36.378141Z [-] CowrieSSHFactory starting on 2222
2026-10-07T20:24:36.383941Z [cowrie.ssh.factory.CowrieSSHFactory#info] Starting factory <cowrie.ssh.factory.CowrieSSHFa
ctory object at 0x7af71aab62a0>
2026-10-07T20:24:36.562684Z [cowrie.ssh.factory.CowrieSSHFactory#info] Ready to accept SSH connections

```

Figure 3: Cowrie log confirming that CowrieSSHFactory is up on port 2222 and "Ready to accept SSH connections".

```

virtualbox-guest-utils is already the newest version (7.0.16-dfsg-2ubuntu1.3).
virtualbox-guest-x11 is already the newest version (7.0.16-dfsg-2ubuntu1.3).
0 upgraded, 0 newly installed, 0 to remove and 42 not upgraded.
oluwatobi@oluwatobi-VirtualBox:/mnt/cdrom$ sudo modprobe vboxguest
oluwatobi@oluwatobi-VirtualBox:/mnt/cdrom$ VBoxClient --clipboard
VBoxClient: error: VbglR3InitUser failed: VERR_FILE_NOT_FOUND
oluwatobi@oluwatobi-VirtualBox:/mnt/cdrom$ ss -lntp | grep 2222
LISTEN 0      50          0.0.0.0:2222      0.0.0.0:*
oluwatobi@oluwatobi-VirtualBox:/mnt/cdrom$

```

Figure 4: Ubuntu host showing a listener on 0.0.0.0:2222 (ss -lntp | grep 2222).

4.2 Connectivity check and Nmap service scan

From Kali, ICMP connectivity to the honeypot host was confirmed with `ping -c 4 10.0.2.15` (4 packets sent, 4 received, 0% loss). An Nmap version-detection scan (`nmap -sV 10.0.2.15`) was then run. It scanned the default 1000 TCP ports in 1.42 seconds, found 999 closed and reported one open port: **2222/tcp, ssh, OpenSSH 9.2p1 Debian 2+deb12u3**. The MAC address was identified as an Oracle VirtualBox virtual NIC.

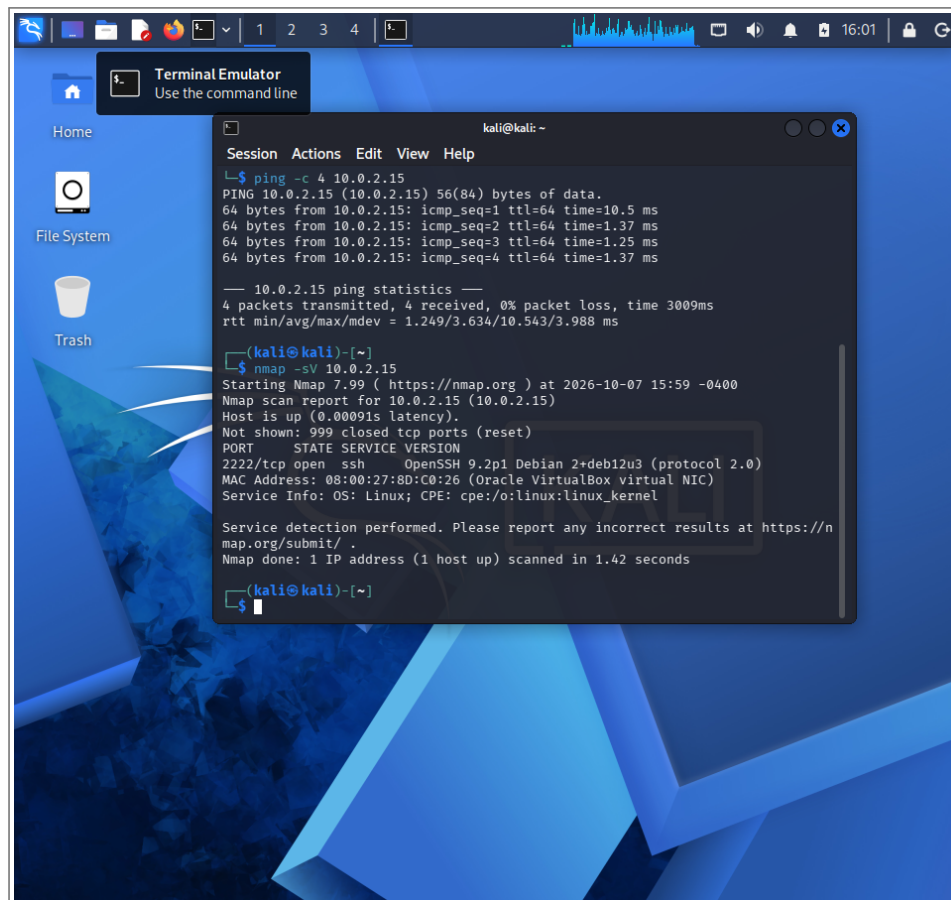


Figure 5: Ping test and Nmap -sV scan results from Kali against 10.0.2.15.

4.3 Simulated unauthorised SSH access

To simulate the post-scan step an attacker would take, an SSH login was made as *root* from the Ubuntu host itself to the honeypot at 127.0.0.1 on port 2222. Cowrie is designed to accept weak or arbitrary credentials so that it can observe what an intruder does next, so the accepted login is expected honeypot behaviour and not a weakness in the host. Cowrie accepted the login and presented an emulated Debian system (hostname *svr04*). Three basic discovery commands were entered: *whoami*, *pwd* and *ls*. The session ended when the emulated shell timed out waiting for input. This was a single manual login; no password-guessing tool was used. The password typed during the test is redacted in the screenshots in this report.

```

Oct 7 20:41
oluwatobi@oluwatobi-VirtualBox: ~
oluwatobi@oluwatobi-VirtualBox:~$ ssh -p 6415 testuser@127.0.0.1
ssh: connect to host 127.0.0.1 port 6415: Connection refused
oluwatobi@oluwatobi-VirtualBox:~$ ss -lntp | grep 6415
oluwatobi@oluwatobi-VirtualBox:~$ grep -n "listen_endpoints" ~/cowrie/etc/cowrie.cfg
grep: /home/oluwatobi/cowrie/etc/cowrie.cfg: No such file or directory
oluwatobi@oluwatobi-VirtualBox:~$ grep -n "listen_endpoints" ~/cowrie/etc/cowrie.cfg
grep: /home/oluwatobi/cowrie/etc/cowrie.cfg: No such file or directory
oluwatobi@oluwatobi-VirtualBox:~$ ssh -p 2222 root@127.0.0.1
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:6sMMDaOTwK7cZS0aZyVnTvRxw1ti0N15noZCQqK70c0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
root@127.0.0.1's password:
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@svr04:~# whoami
root
root@svr04:~# pwd
/root
root@svr04:~# ls
root@svr04:~# timed out waiting for input: auto-logout
Connection to 127.0.0.1 closed.
oluwatobi@oluwatobi-VirtualBox:~$

```

Figure 6: SSH login to the honeypot on port 2222 and the discovery commands entered.

Cowrie logged every step: the new connection, the remote SSH version and HASSH fingerprint, the accepted root login, the shell request, each command and the closing of the TTY log.

```

Oct 7 20:43
cowrie@oluwatobi-VirtualBox: ~/cowrie
2026-10-07T20:24:36.383941Z [cowrie.ssh.factory.CowrieSSHFactory#info] Starting factory <cowrie.ssh.factory.CowrieSSHFactory object at 0x7af71aab62a0>
2026-10-07T20:24:36.562684Z [cowrie.ssh.factory.CowrieSSHFactory#info] Ready to accept SSH connections
2026-10-07T20:34:52.282457Z [cowrie.ssh.factory.CowrieSSHFactory] No moduli, no diffie-hellman-group-exchange-sha1
2026-10-07T20:34:52.298898Z [cowrie.ssh.factory.CowrieSSHFactory] No moduli, no diffie-hellman-group-exchange-sha256
2026-10-07T20:34:52.351216Z [ssh,10d0ac801daa,127.0.0.1] New connection: 127.0.0.1:39834 (127.0.0.1:2222) [session: 10d0ac801daa]
2026-10-07T20:34:52.363025Z [ssh,10d0ac801daa,127.0.0.1] Remote SSH version: SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.19
2026-10-07T20:34:52.412822Z [ssh,10d0ac801daa,127.0.0.1] SSH client hassh fingerprint: aae6b9604f6f3356543709a376d7f657
2026-10-07T20:35:35.327008Z [ssh,10d0ac801daa,127.0.0.1] login attempt [root/[REDACTED]] succeeded
2026-10-07T20:35:35.333413Z [HoneyPotSSTransport,0,127.0.0.1] Initialized emulated server as architecture: linux-x64-lsb
2026-10-07T20:35:35.350770Z [HoneyPotSSTransport,0,127.0.0.1] channel open
2026-10-07T20:35:35.362027Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSTransport,0,127.0.0.1] Handling pty request: b'xterm-256color' (24, 80, 0, 0)
2026-10-07T20:35:35.364068Z [ssh,10d0ac801daa,127.0.0.1] Terminal Size: 80 24
2026-10-07T20:35:35.366418Z [ssh,10d0ac801daa,127.0.0.1] request_env: LANG=en_US.UTF-8
2026-10-07T20:35:35.367277Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSTransport,0,127.0.0.1] Getting shell
2026-10-07T20:36:51.834327Z [ssh,10d0ac801daa,127.0.0.1] CMD: whoami
2026-10-07T20:36:51.883335Z [HoneyPotSSTransport,0,127.0.0.1] Command found: whoami
2026-10-07T20:37:02.228049Z [ssh,10d0ac801daa,127.0.0.1] CMD: pwd
2026-10-07T20:37:02.235768Z [HoneyPotSSTransport,0,127.0.0.1] Command found: pwd
2026-10-07T20:37:07.694484Z [ssh,10d0ac801daa,127.0.0.1] CMD: ls
2026-10-07T20:37:07.700953Z [HoneyPotSSTransport,0,127.0.0.1] Command found: ls
2026-10-07T20:40:07.055171Z [twisted.conch.ssh.session#info] exitCode: 1
2026-10-07T20:40:07.073045Z [ssh,10d0ac801daa,127.0.0.1] Closing TTY Log: var/lib/cowrie/tty/48c28e634a269aa42468dcad54bba62a7d4f9553b245807e495c6a67d2b07e43 after 271703 milliseconds
2026-10-07T20:40:07.075510Z [cowrie.ssh.connection.CowrieSSHConnection#info] sending close 0
2026-10-07T20:40:07.085158Z [HoneyPotSSTransport,0,127.0.0.1] remote close
2026-10-07T20:40:07.090490Z [HoneyPotSSTransport,0,127.0.0.1] Got remote error, code 11 reason: b'disconnected by user'
2026-10-07T20:40:07.098095Z [HoneyPotSSTransport,0,127.0.0.1] avatar root logging out

```

Figure 7: Cowrie log for the local test session (session 10d0ac801daa). Password redacted.

4.4 Network traffic capture with Wireshark

Wireshark was running on the Kali interface *eth0* while the SSH client connected from Kali to the honeypot. The capture contained 47 packets with none dropped. The filter *ip.addr == 10.0.2.15 && tcp.port == 2222* reduced the view to the 17 packets of the SSH session. The other packets in the capture (ARP, a DNS lookup and an HTTP request from the Ubuntu host) are normal background traffic unrelated to the SSH session.

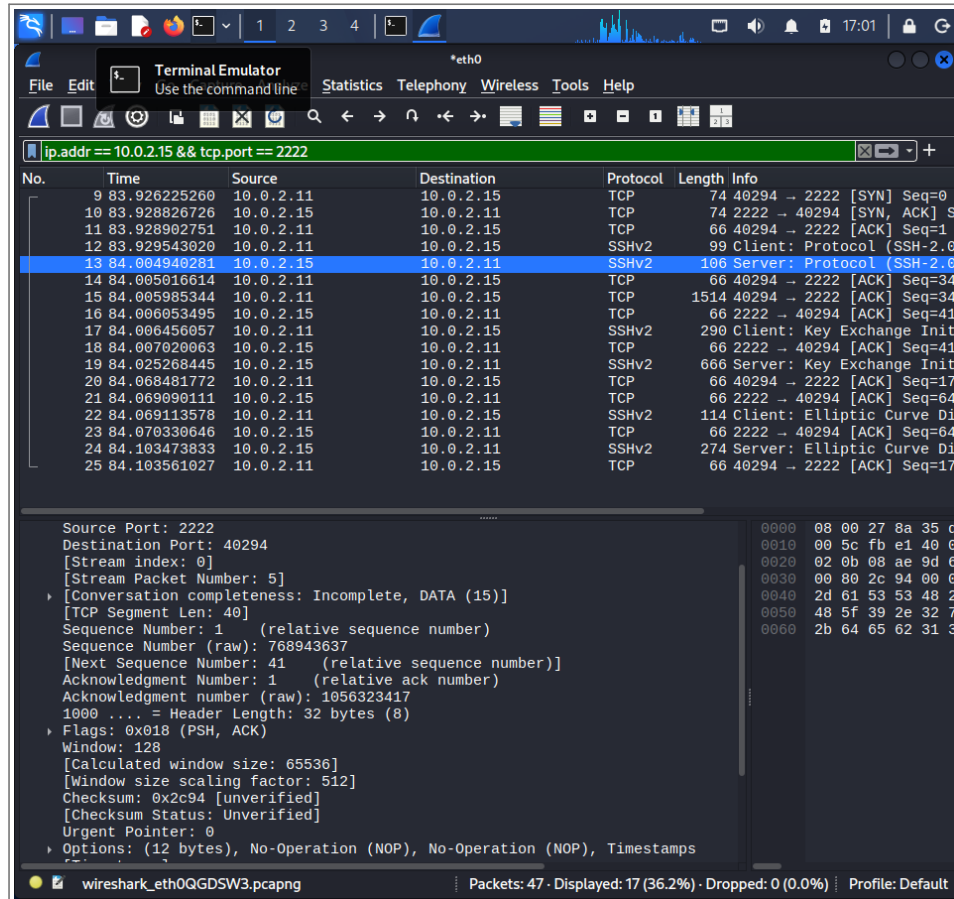


Figure 8: Filtered capture of the SSH session: TCP handshake (packets 9 to 11), SSH protocol banners (12 and 13) and key exchange (17 onwards).

Packets	Observation
9, 10, 11	TCP three-way handshake: SYN from 10.0.2.11:40294, SYN/ACK from 10.0.2.15:2222, then ACK.
12, 13	SSH protocol identification: client banner, then server banner presented by Cowrie.
17, 19	SSH Key Exchange Init from client (290 bytes) and server (666 bytes), negotiating algorithms.
22, 24	Elliptic Curve Diffie-Hellman Key Exchange Init and Reply with New Keys, after which traffic is encrypted.

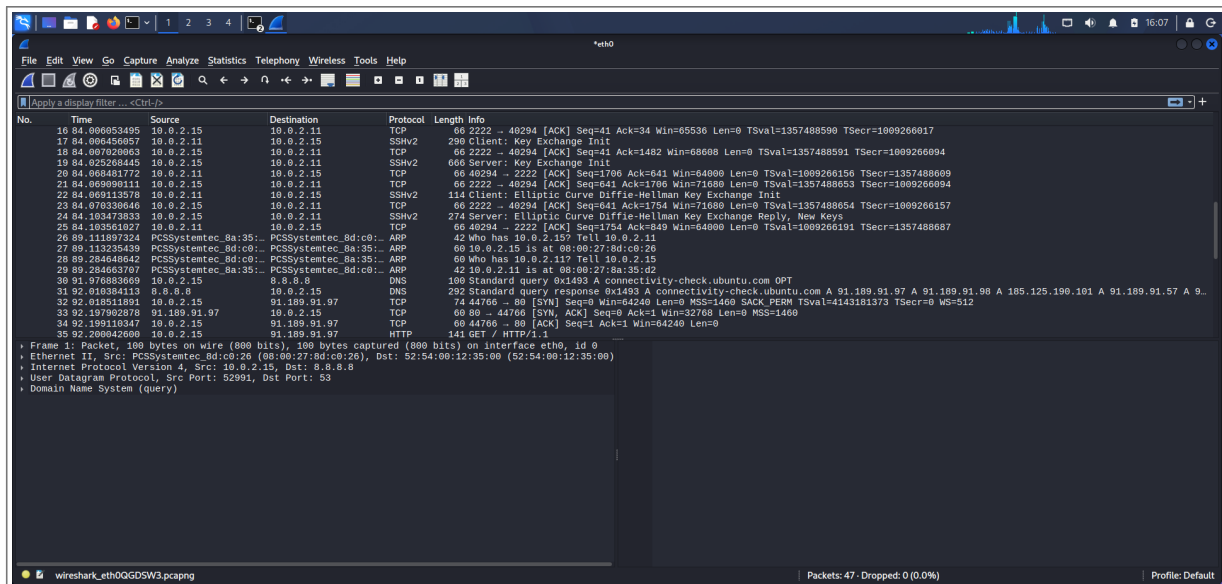


Figure 9: Full Wireshark view: SSH key exchange packets followed by unrelated background ARP, DNS and HTTP traffic.

4.5 Cowrie telemetry for the Kali session

The Cowrie log recorded two connections from 10.0.2.11. The first (session e4b9a9b04144, source port 38578) lasted only 63 milliseconds and is consistent with the Nmap service probe. The second (session 63f7a45ac1bf, source port 40294) is the SSH connection captured in Wireshark. Cowrie logged the client banner *SSH-2.0-OpenSSH_10.4p1 Debian-4* and the HASSH fingerprint, then closed the session when its timeout was reached after 123,340 ms. No login was recorded for this session.

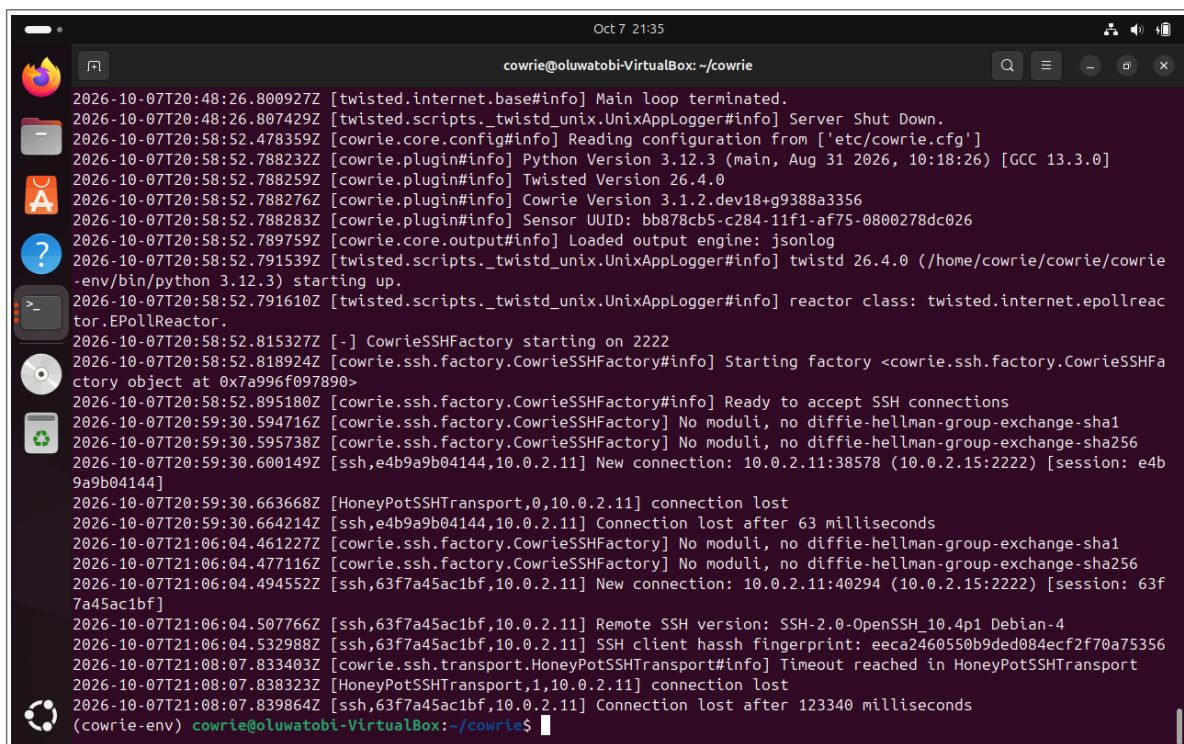


Figure 10: Cowrie log entries for the Nmap probe and the Kali SSH session.

The JSON log (*cowrie.json*) holds the same session in structured form, including source and destination IP and ports, key-exchange algorithms, ciphers and the session-closed event. Cowrie also stores TTY recordings under *var/lib/cowrie/tty/*. The recording found there (about 1.5 KB, identified by *file* as binary data) was created by the earlier local test session, whose log shows it being closed at 20:40:07.

```

Oct 7 21:51
cowrie@oluwatobi-VirtualBox: ~/cowrie

{"@openssh.com","aes128-gcm@openssh.com","aes256-gcm@openssh.com","aes128-ctr","aes192-ctr","aes256-ctr"],"macCS":["umac-64-etm@openssh.com","umac-128-etm@openssh.com","hmac-sha2-256-etm@openssh.com","hmac-sha2-512-etm@openssh.com","hmac-sha1-etm@openssh.com","umac-64@openssh.com","umac-128@openssh.com","hmac-sha2-256","hmac-sha2-512","hmac-sha1"],"compCS":["none"],"zlib@openssh.com"],"langCS":[""],"eventid":"cowrie.client.kex","sensor":"oluwatobi-VirtualBox","uuid":"bb878cb5-c284-11f1-af75-0800278dc026","timestamp":"2026-10-07T21:06:04.532178Z","message":"SSH client hassh fingerprint: eeca2460550b9ded084ecf2f70a75356"}
{"session":"63f7a45ac1bf","protocol":"ssh","src_ip":"10.0.2.11","src_port":40294,"dst_ip":"10.0.2.15","dst_port":2222,"duration_ms":123340,"eventid":"cowrie.session.closed","sensor":"oluwatobi-VirtualBox","uuid":"bb878cb5-c284-11f1-af75-0800278dc026","timestamp":"2026-10-07T21:08:07.838857Z","message":"Connection lost after 123340 milliseconds"}
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ find var/lib/cowrie -maxdepth 2 -type f | head -30
var/lib/cowrie/ssh_host_ed25519_key
var/lib/cowrie/downloads/.gitignore
var/lib/cowrie/ssh_host_ecdsa_key.pub
var/lib/cowrie/.gitignore
var/lib/cowrie/ssh_host_rsa_key
var/lib/cowrie/snapshots/.gitignore
var/lib/cowrie/ssh_host_ed25519_key.pub
var/lib/cowrie/ssh_host_ecdsa_key
var/lib/cowrie/uuid
var/lib/cowrie/tty/.gitignore
var/lib/cowrie/tty/48c28e634a269aa42468dcad54bba62a7d4f9553b245807e495c6a67d2b07e43
var/lib/cowrie/ssh_host_rsa_key.pub
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ ls -lh var/lib/cowrie/tty/
total 4.0K
-rw-r--r-- 1 cowrie cowrie 1.5K Oct  7 20:40 48c28e634a269aa42468dcad54bba62a7d4f9553b245807e495c6a67d2b07e43
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ filr var/lib/cowrie/tty/48c28e634a269aa42468dcad54bba62a7d4f9553b245807e495c6a67d2b07e43
Command 'filr' not found, did you mean:
  command 'file' from deb file (1:5.45-2)
Try: apt install <deb name>
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ file var/lib/cowrie/tty/48c28e634a269aa42468dcad54bba62a7d4f9553b245807e495c6a67d2b07e43
var/lib/cowrie/tty/48c28e634a269aa42468dcad54bba62a7d4f9553b245807e495c6a67d2b07e43: data

```

Figure 11: Structured JSON session record and the TTY recording stored under var/lib/cowrie/tty/.

5. Findings and Analysis

5.1 Correlation of network and honeypot evidence

Attribute	Wireshark (network)	Cowrie (host)	Match
Source IP	10.0.2.11	10.0.2.11	Yes
Source port	40294	40294	Yes
Destination	10.0.2.15:2222	10.0.2.15:2222	Yes
Protocol stage	Handshake, banners, key exchange	SSH version and HASSH logged	Yes
Service identity	Nmap: OpenSSH 9.2p1 Debian 2+deb12u3	Emulated Debian OpenSSH banner	Yes

5.2 Key observations

- **Detection.** Any connection to the honeypot is suspicious by definition, because no legitimate service depends on it. Both the scan and the SSH session were recorded without any extra tooling.
- **Deception works.** Nmap reported OpenSSH 9.2p1 on Debian 12, and the shell prompt identified a Debian system named svr04. The real host is Ubuntu 24.04. The scan result therefore describes the honeypot's disguise, not the underlying machine.
- **Encryption limits packet analysis.** Wireshark shows addresses, ports, timing, banners and key-exchange details, but after key exchange the content is encrypted. Commands and credentials were only visible because Cowrie terminates the SSH session itself and logs what happens inside it.
- **HASSH is a useful indicator.** The fingerprint eeca2460550b9ded084ecf2f70a75356 identifies the Kali SSH client's negotiation behaviour. The local test client had a different value (aae6b9604f6f3356543709a376d7f657), showing that HASSH distinguishes client software.
- **Behaviour matches early attacker activity.** Scanning for services, logging in over SSH and running whoami, pwd and ls are typical first steps after gaining access.

6. MITRE ATT&CK Mapping

Technique	ID	Evidence in this lab
Network Service Discovery	T1046	Nmap -sV scan finding 2222/tcp open
Remote Services: SSH	T1021.004	SSH login to the honeypot on port 2222
System Owner/User Discovery	T1033	<i>whoami</i> run in the emulated shell
File and Directory Discovery	T1083	<i>ls</i> run in the emulated shell

The mapping is indicative. The activity was a controlled simulation, so these techniques show what the telemetry would look like in a real incident rather than confirming one.

7. Recommendations

- **Centralised logging.** Forward Cowrie and network logs to a SIEM such as Wazuh or Splunk for correlation.
- **Alerting.** Raise an alert on any connection to a honeypot address, on repeated connection attempts and on new HASSH values.
- **Network monitoring.** Watch for scans and unexpected SSH connections to administrative ports from non-management hosts.
- **Segmentation.** Keep honeypots isolated from production networks so they cannot be used for lateral movement.
- **Evidence preservation.** Keep packet captures, JSON logs and TTY recordings together so network and host evidence can be matched later.
- **Hardening real systems.** Restrict SSH to trusted source addresses, use key-based authentication and disable root login.

8. Limitations

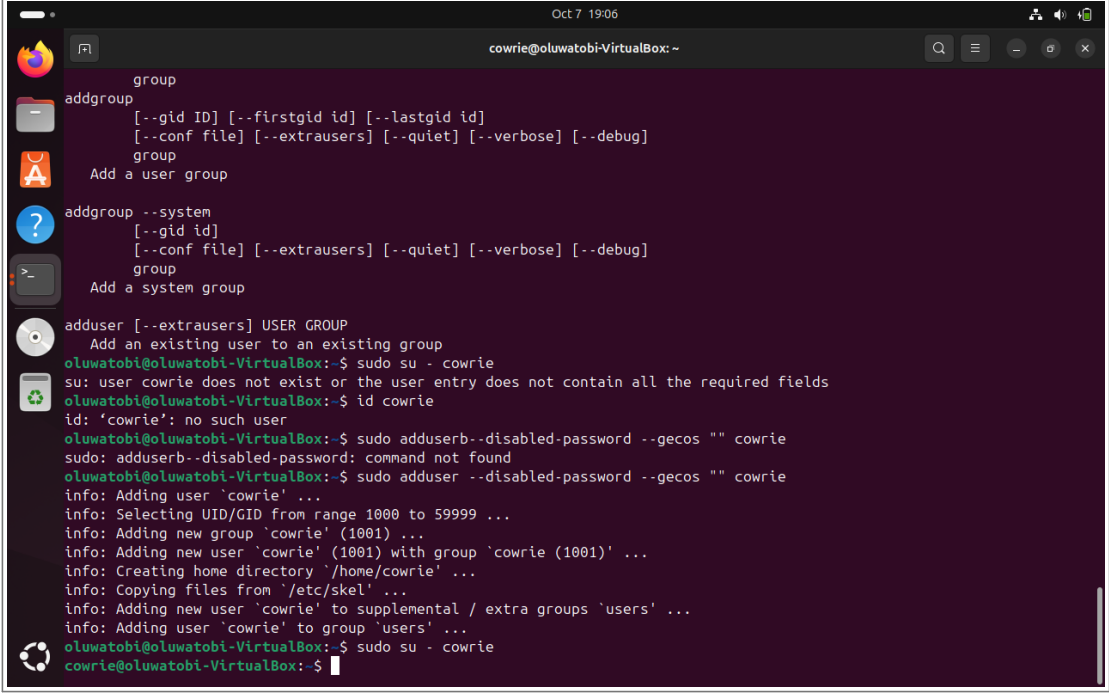
- The exercise took place in an isolated VirtualBox lab. It shows detection workflow, not a real-world compromise.
- Only a version-detection scan and a single manual SSH login were performed. No brute-force or exploitation tools were used.
- The Kali session recorded in Wireshark stopped after key exchange and did not log in, so the commands in Section 4.3 belong to the separate login made from the Ubuntu host to 127.0.0.1.
- The Nmap scan used the default setting, which covers the 1000 most common TCP ports. A full-range scan (-p-) was not run.
- The VM clocks were not synchronised, so the Kali and Cowrie timestamps do not line up exactly. Correlation relies on IP address, source port and session behaviour instead.
- SSH traffic after key exchange is encrypted and cannot be read in Wireshark.

9. Conclusion

The lab met the assignment aims. Cowrie was deployed and shown running, Nmap identified the exposed service, Wireshark captured the handshake and key exchange, and Cowrie logged the same connection with source details, client fingerprint, duration and a TTY artefact. Combining network-level and host-level evidence gives an analyst a much clearer picture than either source alone.

Appendix A: Additional Setup Screenshots

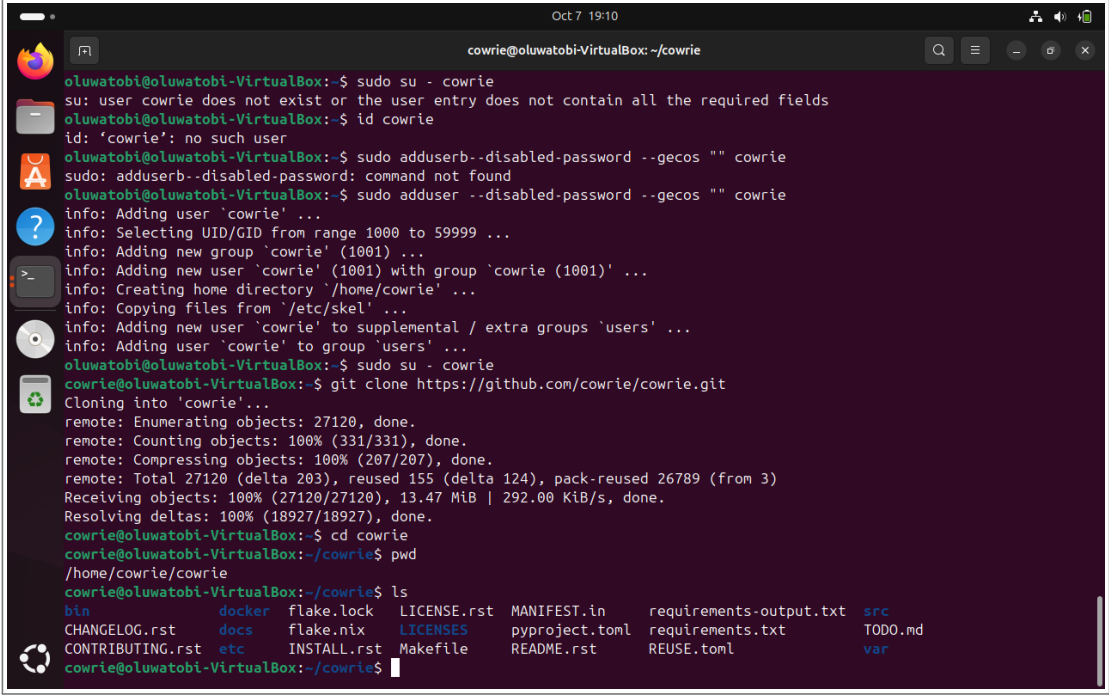
These screenshots show the preparation steps. Some include typing mistakes and the corrections that followed, which are left as captured.



```

Oct 7 19:06
cowrie@oluwatobi-VirtualBox: ~
group
addgroup
  [--gid ID] [--firstgid id] [--lastgid id]
  [--conf file] [--extrausers] [--quiet] [--verbose] [--debug]
group
Add a user group
addgroup --system
  [--gid id]
  [--conf file] [--extrausers] [--quiet] [--verbose] [--debug]
group
Add a system group
adduser [--extrausers] USER GROUP
Add an existing user to an existing group
oluwatobi@oluwatobi-VirtualBox:~$ sudo su - cowrie
su: user cowrie does not exist or the user entry does not contain all the required fields
oluwatobi@oluwatobi-VirtualBox:~$ id cowrie
id: 'cowrie': no such user
oluwatobi@oluwatobi-VirtualBox:~$ sudo adduserb--disabled-password --gecos "" cowrie
sudo: adduserb--disabled-password: command not found
oluwatobi@oluwatobi-VirtualBox:~$ sudo adduser --disabled-password --gecos "" cowrie
info: Adding user 'cowrie' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'cowrie' (1001) ...
info: Adding new user 'cowrie' (1001) with group 'cowrie (1001)' ...
info: Creating home directory '/home/cowrie' ...
info: Copying files from '/etc/skel' ...
info: Adding new user 'cowrie' to supplemental / extra groups 'users' ...
info: Adding user 'cowrie' to group 'users' ...
oluwatobi@oluwatobi-VirtualBox:~$ sudo su - cowrie
cowrie@oluwatobi-VirtualBox:~$
  
```

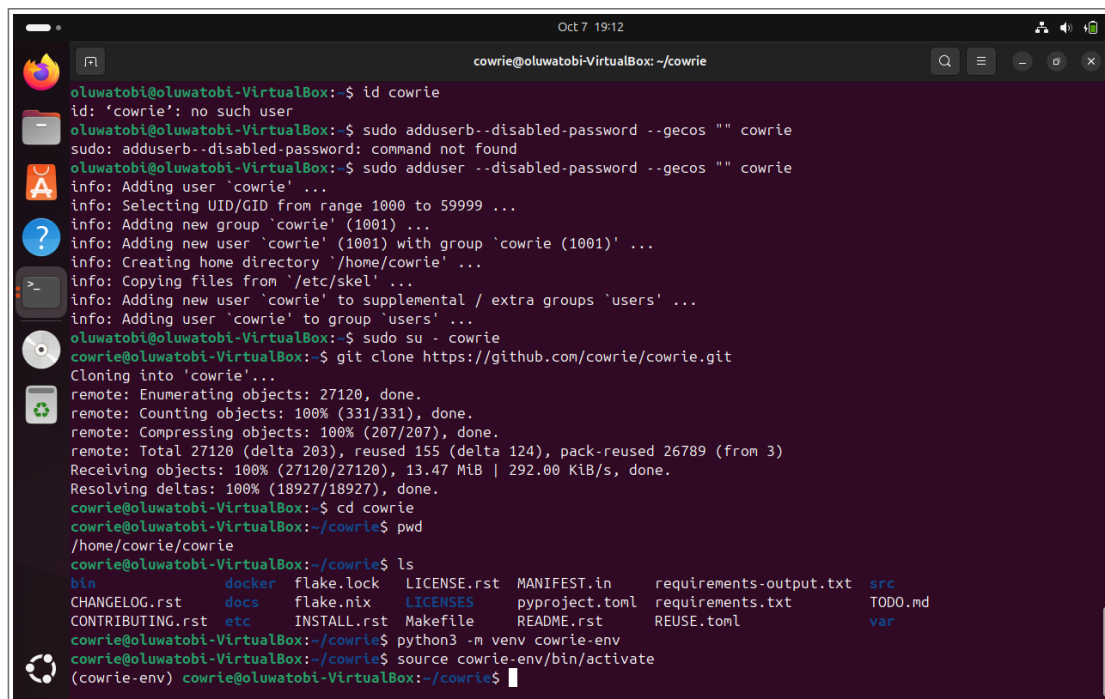
Figure A1: Creation of the dedicated cowrie user.



```

Oct 7 19:10
cowrie@oluwatobi-VirtualBox: ~/cowrie
oluwatobi@oluwatobi-VirtualBox:~$ sudo su - cowrie
su: user cowrie does not exist or the user entry does not contain all the required fields
oluwatobi@oluwatobi-VirtualBox:~$ id cowrie
id: 'cowrie': no such user
oluwatobi@oluwatobi-VirtualBox:~$ sudo adduserb--disabled-password --gecos "" cowrie
sudo: adduserb--disabled-password: command not found
oluwatobi@oluwatobi-VirtualBox:~$ sudo adduser --disabled-password --gecos "" cowrie
info: Adding user 'cowrie' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'cowrie' (1001) ...
info: Adding new user 'cowrie' (1001) with group 'cowrie (1001)' ...
info: Creating home directory '/home/cowrie' ...
info: Copying files from '/etc/skel' ...
info: Adding new user 'cowrie' to supplemental / extra groups 'users' ...
info: Adding user 'cowrie' to group 'users' ...
oluwatobi@oluwatobi-VirtualBox:~$ sudo su - cowrie
cowrie@oluwatobi-VirtualBox:~$ git clone https://github.com/cowrie/cowrie.git
Cloning into 'cowrie'...
remote: Enumerating objects: 27120, done.
remote: Counting objects: 100% (331/331), done.
remote: Compressing objects: 100% (207/207), done.
remote: Total 27120 (delta 203), reused 155 (delta 124), pack-reused 26789 (from 3)
Receiving objects: 100% (27120/27120), 13.47 MiB | 292.00 KiB/s, done.
Resolving deltas: 100% (18927/18927), done.
cowrie@oluwatobi-VirtualBox:~$ cd cowrie
cowrie@oluwatobi-VirtualBox:~/cowrie$ pwd
/home/cowrie/cowrie
cowrie@oluwatobi-VirtualBox:~/cowrie$ ls
bin          docker  flake.lock  LICENSE.rst  MANIFEST.in  requirements-output.txt  src
CHANGELOG.rst  docs   flake.nix   LICENSES     pyproject.toml  requirements.txt         TODO.md
CONTRIBUTING.rst  etc    INSTALL.rst  Makefile     README.rst     REUSE.toml              var
  
```

Figure A2: Cowrie repository cloned from GitHub into /home/cowrie/cowrie.



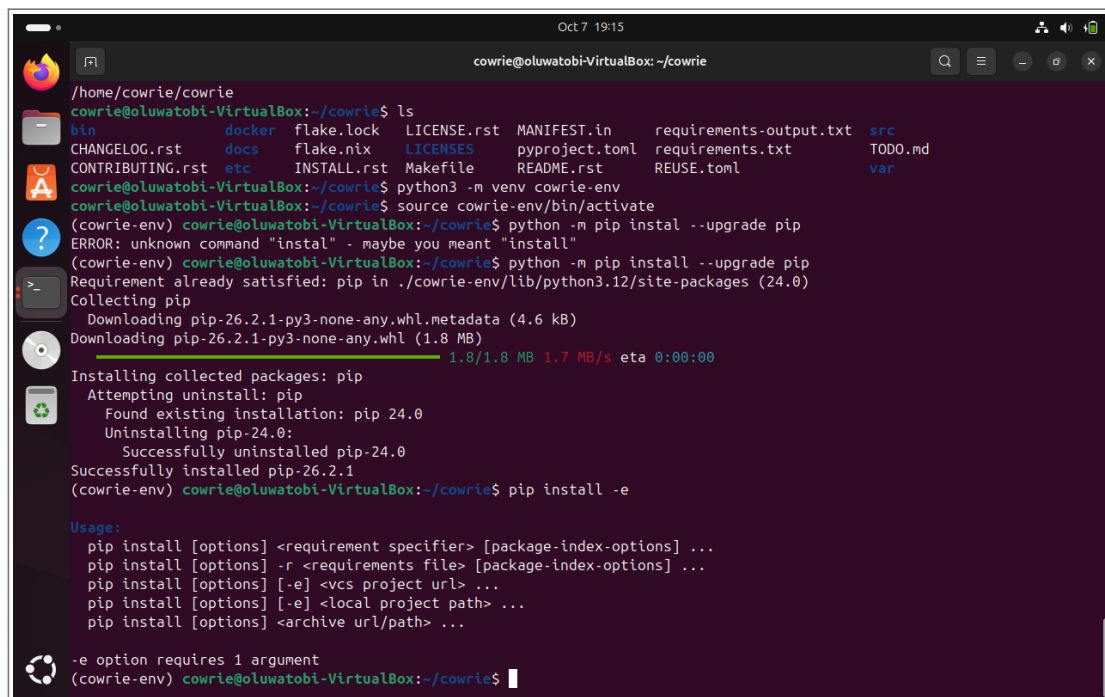
```

Oct 7 19:12
cowrie@oluwatobi-VirtualBox: ~/cowrie

oluwatobi@oluwatobi-VirtualBox:~$ id cowrie
id: 'cowrie': no such user
oluwatobi@oluwatobi-VirtualBox:~$ sudo adduserb--disabled-password --gecos "" cowrie
sudo: adduserb--disabled-password: command not found
oluwatobi@oluwatobi-VirtualBox:~$ sudo adduser --disabled-password --gecos "" cowrie
info: Adding user 'cowrie' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'cowrie' (1001) ...
info: Adding new user 'cowrie' (1001) with group 'cowrie' (1001) ...
info: Creating home directory '/home/cowrie' ...
info: Copying files from '/etc/skel' ...
info: Adding new user 'cowrie' to supplemental / extra groups 'users' ...
info: Adding user 'cowrie' to group 'users' ...
oluwatobi@oluwatobi-VirtualBox:~$ sudo su - cowrie
cowrie@oluwatobi-VirtualBox:~$ git clone https://github.com/cowrie/cowrie.git
Cloning into 'cowrie'...
remote: Enumerating objects: 27120, done.
remote: Counting objects: 100% (331/331), done.
remote: Compressing objects: 100% (207/207), done.
remote: Total 27120 (delta 203), reused 155 (delta 124), pack-reused 26789 (from 3)
Receiving objects: 100% (27120/27120), 13.47 MiB | 292.00 KiB/s, done.
Resolving deltas: 100% (18927/18927), done.
cowrie@oluwatobi-VirtualBox:~$ cd cowrie
cowrie@oluwatobi-VirtualBox:~/cowrie$ pwd
/home/cowrie/cowrie
cowrie@oluwatobi-VirtualBox:~/cowrie$ ls
bin          docker      flake.lock  LICENSE.rst  MANIFEST.in  requirements-output.txt  src
CHANGELOG.rst  docs       flake.nix   LICENSES     pyproject.toml  requirements.txt         TODO.md
CONTRIBUTING.rst  etc       INSTALL.rst  Makefile     README.rst     REUSE.toml              var
cowrie@oluwatobi-VirtualBox:~/cowrie$ python3 -m venv cowrie-env
cowrie@oluwatobi-VirtualBox:~/cowrie$ source cowrie-env/bin/activate
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$

```

Figure A3: Python virtual environment cowrie-env created and activated.



```

Oct 7 19:15
cowrie@oluwatobi-VirtualBox: ~/cowrie

/home/cowrie/cowrie
cowrie@oluwatobi-VirtualBox:~/cowrie$ ls
bin          docker      flake.lock  LICENSE.rst  MANIFEST.in  requirements-output.txt  src
CHANGELOG.rst  docs       flake.nix   LICENSES     pyproject.toml  requirements.txt         TODO.md
CONTRIBUTING.rst  etc       INSTALL.rst  Makefile     README.rst     REUSE.toml              var
cowrie@oluwatobi-VirtualBox:~/cowrie$ python3 -m venv cowrie-env
cowrie@oluwatobi-VirtualBox:~/cowrie$ source cowrie-env/bin/activate
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ python -m pip install --upgrade pip
ERROR: unknown command "instal" - maybe you meant "install"
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ python -m pip install --upgrade pip
Requirement already satisfied: pip in ./cowrie-env/lib/python3.12/site-packages (24.0)
Collecting pip
  Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
  Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 1.7 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.0
    Uninstalling pip-24.0:
      Successfully uninstalled pip-24.0
  Successfully installed pip-26.2.1
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$ pip install -e

Usage:
  pip install [options] <requirement specifier> [package-index-options] ...
  pip install [options] -r <requirements file> [package-index-options] ...
  pip install [options] [-e] <vcs project url> ...
  pip install [options] [-e] <local project path> ...
  pip install [options] <archive url/path> ...

-e option requires 1 argument
(cowrie-env) cowrie@oluwatobi-VirtualBox:~/cowrie$

```

Figure A4: pip upgraded inside the virtual environment.

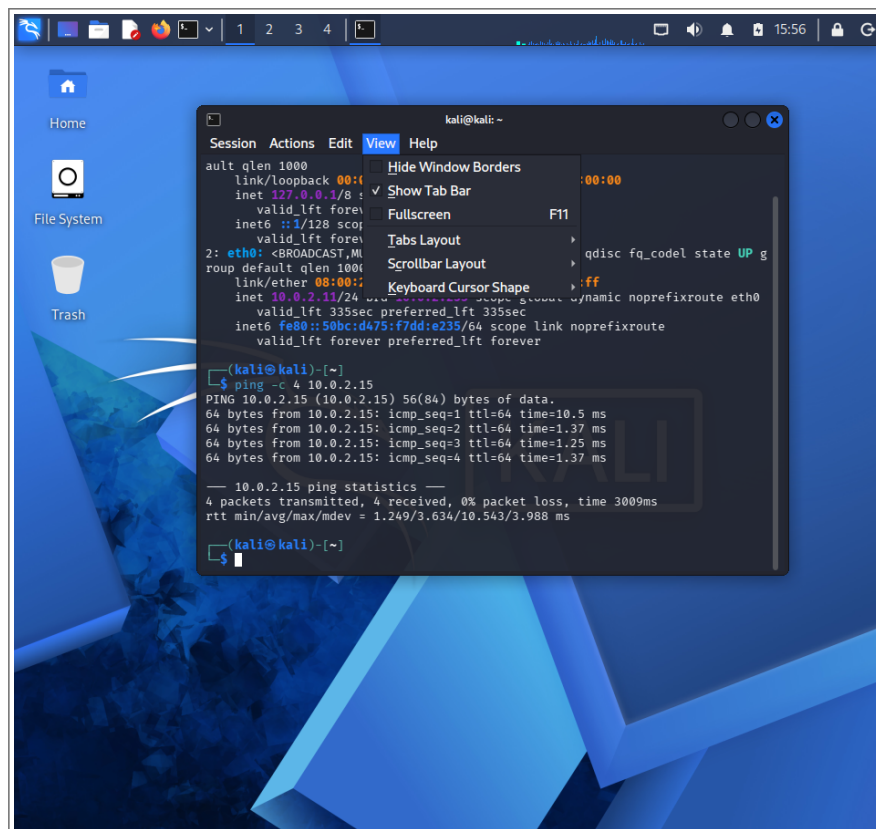


Figure A5: Kali network interface (10.0.2.11) and successful ping to 10.0.2.15.

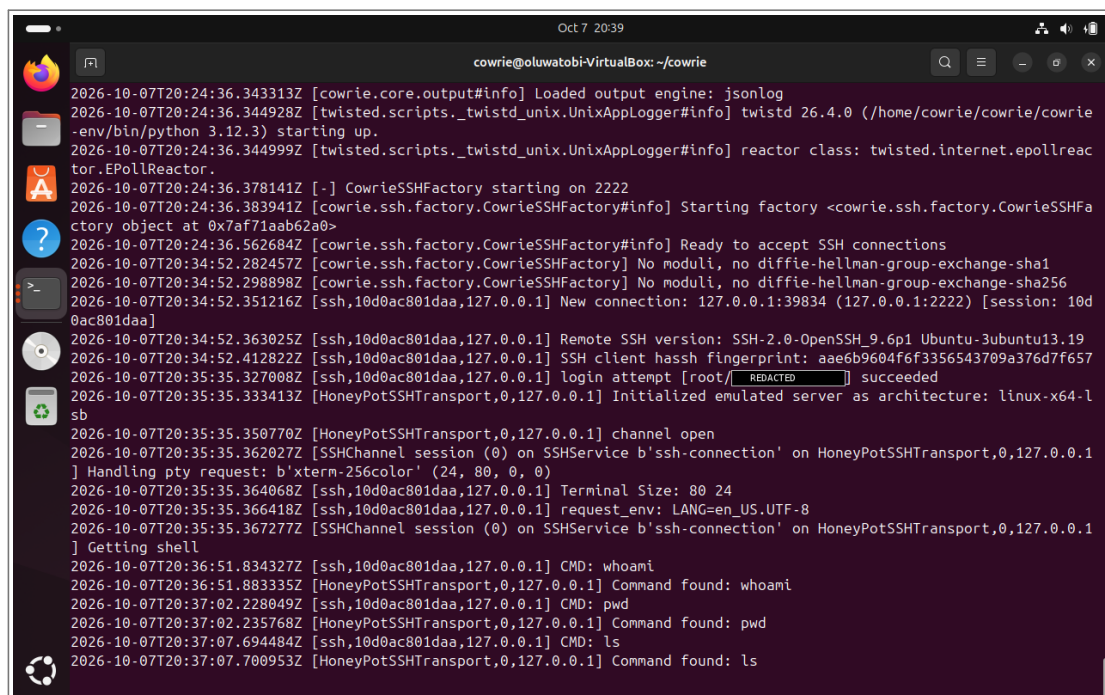


Figure A6: Earlier view of the Cowrie log for the local test session. Password redacted.

Appendix B: Assignment Deliverables Checklist

Required deliverable	Where it is in this report
Kali Linux and Cowrie on separate VMs	Section 3 (Lab Environment) and Appendix A
Screenshot of honeypot running	Figures 2, 3 and 4
Nmap scan results	Figure 5 (Section 4.2)
Wireshark capture (.pcap)	Figures 8 and 9 (screenshots of the capture, as the tutor advised). The capture file wireshark_eth0QGDSW3.pcapng is held on the Kali VM.
1-page incident report	Page 2